

Mohit Kumar Wankhade

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Summary

AI/ML and Software Engineer focused on building production-ready GenAI applications, RAG systems, backend APIs, and machine learning solutions. Experienced with FastAPI, LLM APIs, LangChain, vector search, NLP, deep learning, databases, Dockerized deployments, monitoring, and cloud-based application delivery.

Technical Skills

Programming: Python, C++, JavaScript, SQL

Generative AI: LLMs, RAG, Prompt Engineering, Embeddings, Vector Search, LangChain, Sentence Transformers

Retrieval: FAISS, BM25, Hybrid Search, Cross-Encoder Reranking, Source Attribution, Context Building

Backend/SWE: FastAPI, REST APIs, JWT Authentication, API Design, OOP, Debugging, Logging, Testing

Frontend: React, Vite, Tailwind CSS, Axios, React Router, HTML, CSS

Databases & Cache: MongoDB, MongoDB Atlas, Redis, MySQL

ML/DL: PyTorch, TensorFlow, Scikit-learn, NLP, Computer Vision, Transformers, Multimodal Learning, Classification

DevOps/Cloud: Docker, Docker Compose, Nginx, Prometheus, Grafana, AWS, Oracle Cloud, Git, GitHub

Experience

Tata Consultancy Services (TCS)

May 2025 – Present

Software Engineer – GenAI Applications

Pune, Maharashtra

- Worked on GenAI chatbot workflows involving LLM APIs, semantic retrieval, REST APIs, and backend services for contextual query handling.
- Designed FastAPI-based API modules with request validation, exception handling, structured logging, and reusable service-layer logic.
- Integrated MongoDB-based conversation memory to support context-aware interactions, persistent chat history, and multi-turn conversations.
- Performed monitoring, API testing, debugging, incident analysis, and performance validation to improve application stability and response reliability.

Indian Institute of Technology (IIT) Patna

Jan 2024 – May 2024

NLP Research Intern under Dr. Sriparna Saha

Patna, Bihar

- Developed a multimodal financial complaint identification framework using text, image, sentiment, emotion, severity, and aspect-level signals.
- Worked on financial social media data covering 6,439 annotated samples, 5 financial domains, and 10 aspect categories.
- Built transformer and deep learning pipelines using RoBERTa, ResNet-152, BiGRU, self-attention, and BART, achieving up to 93% complaint classification accuracy.

Projects

MW GenAI Chatbot – Production RAG Assistant

Live Demo | GitHub

FastAPI, React, LangChain, FAISS, BM25, MongoDB Atlas, Redis, Docker, Nginx, Oracle Cloud

- Built and deployed a full-stack RAG chatbot supporting PDF upload, GitHub repository indexing, streaming responses, and source-grounded answers.
- Implemented hybrid retrieval using FAISS dense search, BM25 sparse search, result fusion, and Cross-Encoder reranking to improve context relevance.
- Added JWT authentication, protected routes, MongoDB conversation memory, Redis response caching, and source attribution for PDF pages and GitHub files.
- Containerized and deployed the complete stack using Docker Compose, Nginx reverse proxy, Prometheus metrics, Grafana dashboards, health checks, and Oracle Cloud VM.

Multimodal Financial Complaint Identification Framework

Python, PyTorch, RoBERTa, ResNet-152, BiGRU, BART, NLP, Computer Vision

- Designed an aspect-based multimodal complaint identification framework using textual, visual, sentiment, emotion, and severity features.
- Processed 6,439 financial social media samples across 5 financial domains and 10 aspect categories with complaint/non-complaint annotations.
- Combined RoBERTa text embeddings with ResNet-152 image features and self-attention layers, achieving 93% accuracy/micro-F1 and 91% macro-F1.

Education

Madhav Institute of Technology and Science, Gwalior

2020 – 2024

Bachelor of Technology in Artificial Intelligence and Robotics

CGPA: 8.07/10

Achievements

- GATE Qualified in Data Science & AI and Computer Science
- Solved 440+ problems on LeetCode with 365+ active days and multiple coding badges
- Certifications: Supervised Machine Learning and Advanced Learning Algorithms – Coursera, Stanford University